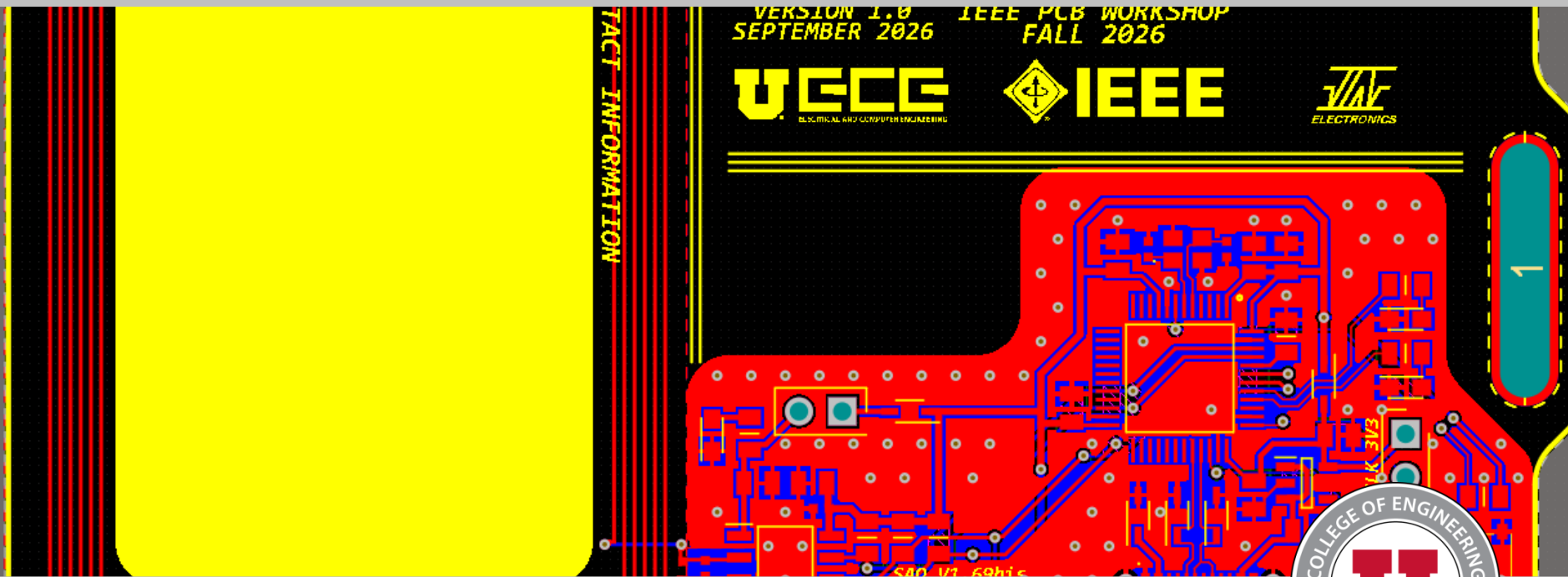


IEEE PCB Design Workshop: (Week 03) Introduction to PCB Layout



Adrian Suchayo and IEEE at the University of Utah
Department of Electrical and Computer Engineering
University of Utah, Salt Lake City, UT



Workshop Outline

Tentatively scheduled for 6-7 weeks

- Sept. 15 – Introduction to Schematics (WEB L102)
- Sept. 22 – Schematics Continued (WEB L112)
- **Sept. 29 – Introduction to PCB Layout (WEB L102)**
- Oct. 6 – Layout Continued (WEB L102)
- Oct. 8 – PCBs DUE!
- **** FALL BREAK ****
- TBD – Soldering Week 1
- TBD – Soldering Week 2
- TBD – Soldering Week 3 (Optional)

Announcements

- Alternative Projects
 - We will be able to get alternative project boards manufactured if submitted by the deadline.
 - Limited to the 10 cm x 10 cm dimensions outlined by JLCPCB.
 - Talk or email me if you have any questions!
- PCB Submission Deadlines
 - For both workshop and custom PCBs, the deadline for submission is **Oct 8th**

Want more experience?

- Consider joining the FSAE tractive team!
 - The Tractive Team is currently looking for students to assist with designing and assembling the electrical system for an electric formula-style race car!
 - No experience required!

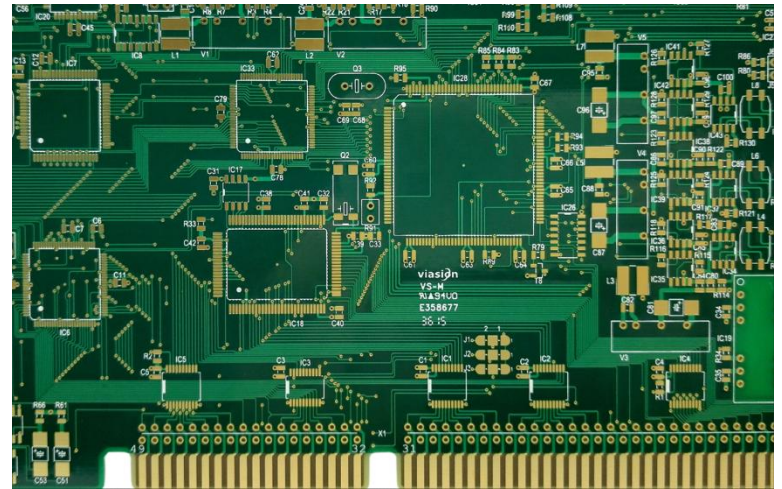
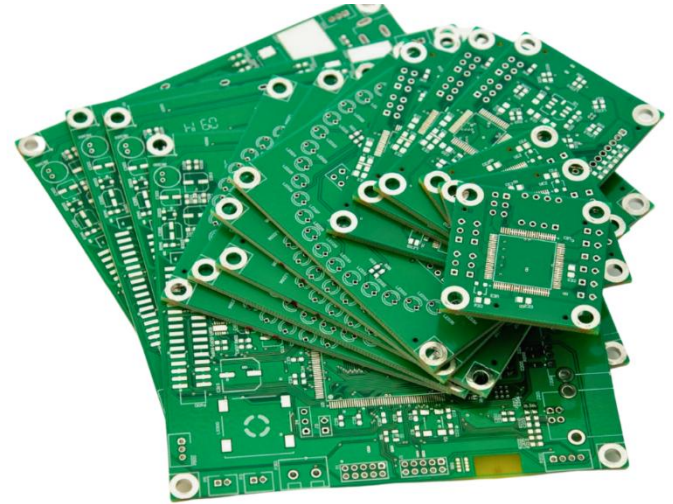


U of U FSAE Discord Link



Printed Circuit Boards

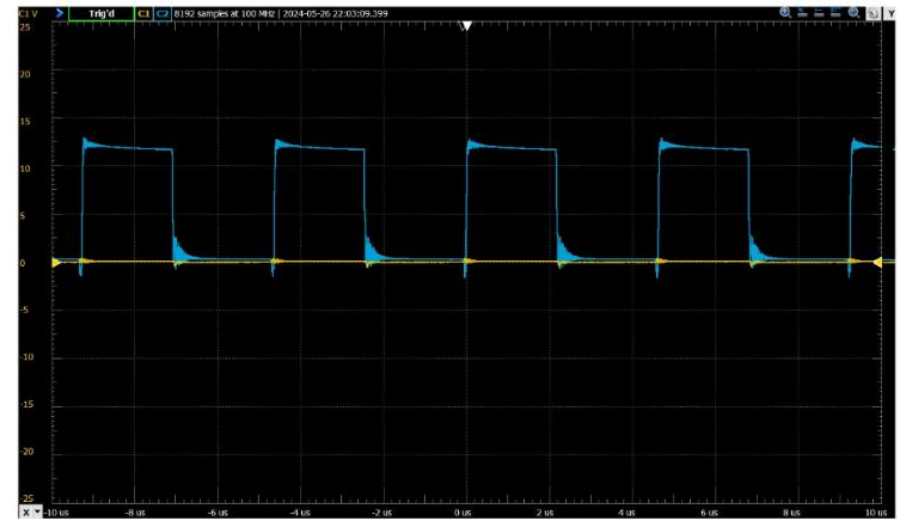
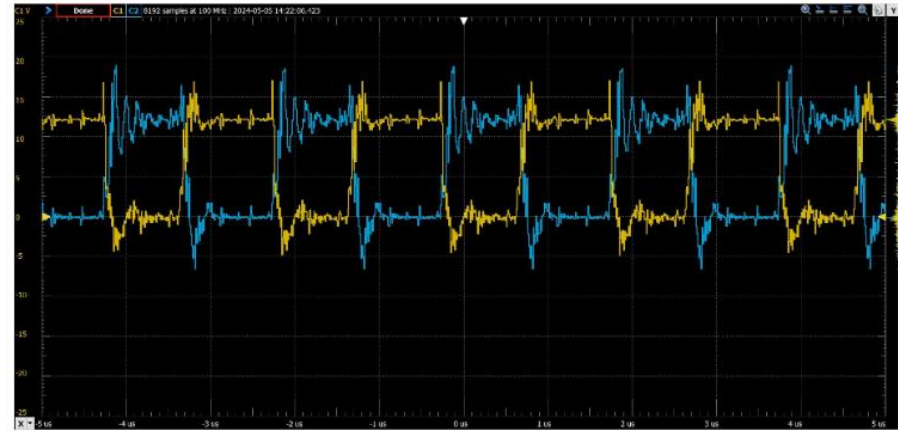
- PCBs: A stack of conductive and non-conductive layers used to connect electrical components in a circuit
- Most common type of electronics assembly
- Components can be soldered to one or both sides
 - Increased Density at Increased Cost for Assembly
- Cheap to produce and manufacture



<https://www.viasion.com/blog/pcb-layers-traces-and-pads-explained/>

Why Use PCBs?

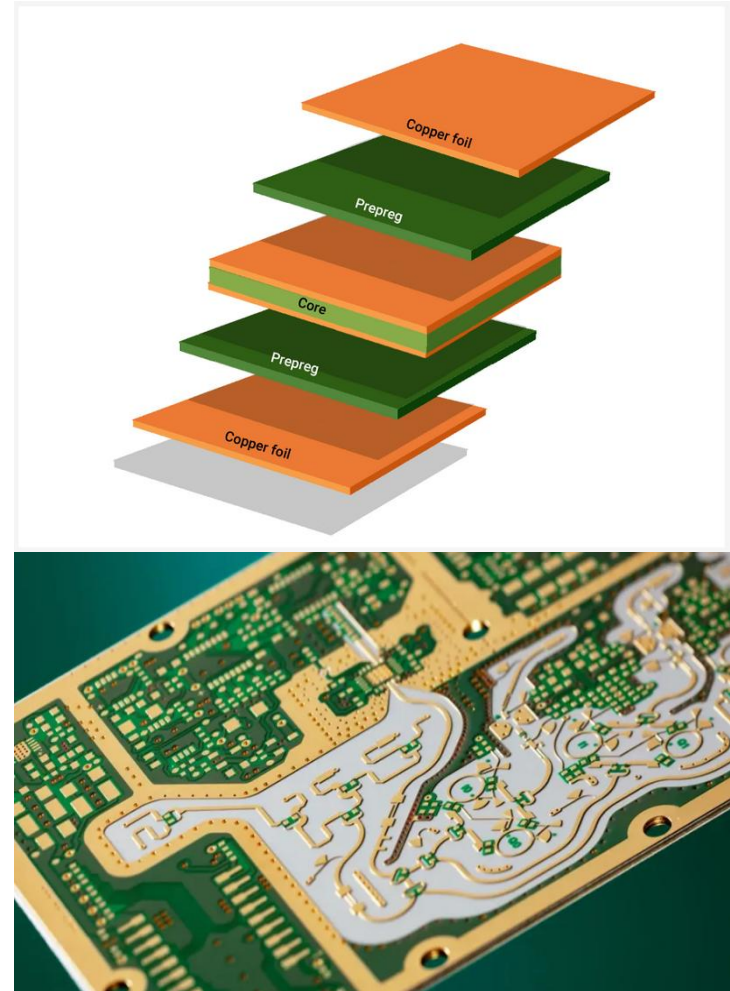
- Standard for most manufacturing processes
- Higher performance compared to breadboard or perfboard circuits
 - Less parasitic capacitances and inductances
- Relatively Durable
- Cheap to produce and manufacture
- High Density



Nick Howard – Tesla Coil Ringing Demo, Adapted from SP25 Workshop

PCB Composition

- PCBs consist of many layers of insulators and conductors
 - Specific configurations is called the “Stackup”
- PCBs can be classified by the number of layers
 - 1-Layer, 2-Layer, 4-Layer, etc.
- Specific applications have tighter design tolerances
 - Ex: Rogers for RF/High Speed

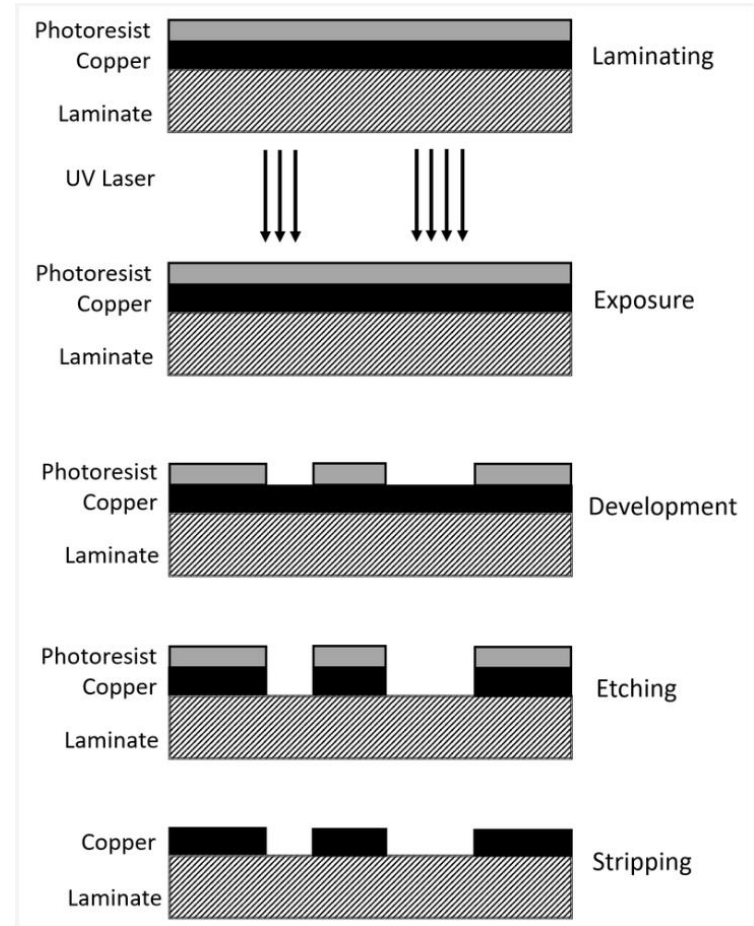
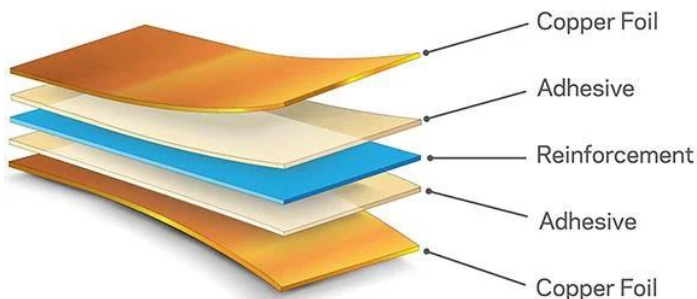


<https://www.protoexpress.com/kb/pcb-manufacturing-overview/>

<https://www.viasion.com/blog/rogers-pcb-properties-materials-advantages-and-applications/>

PCB Composition – Copper and Core

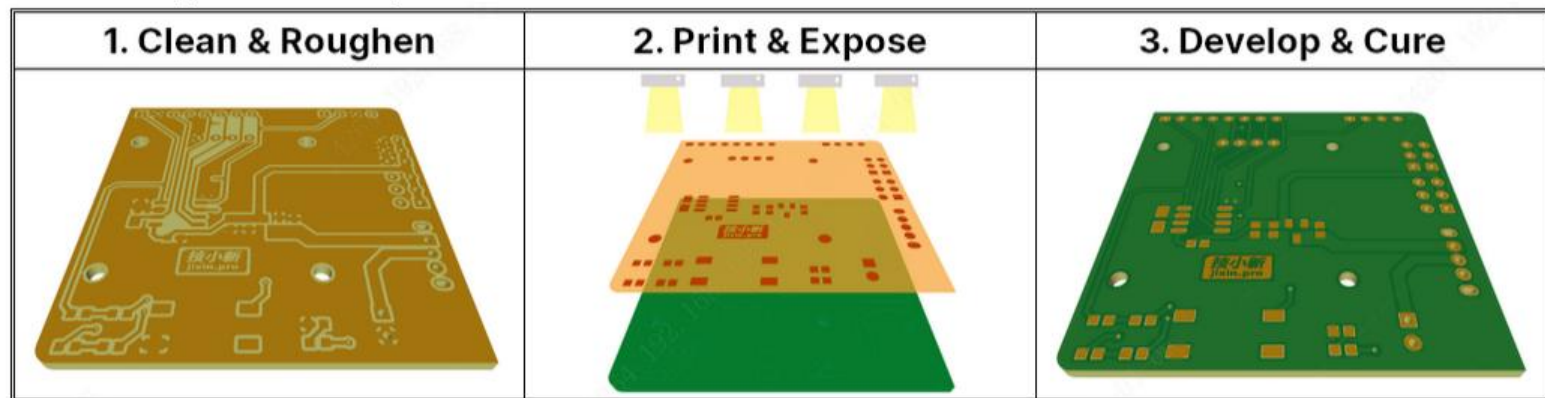
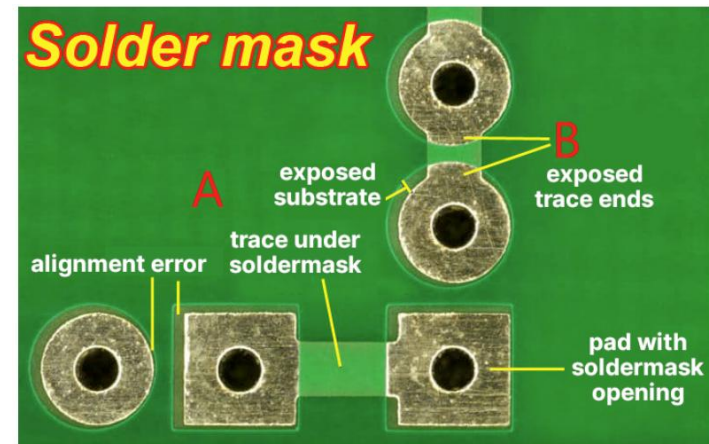
- Printed Circuit Boards (PCBs) typically begin as copper clad boards
 - Copper foil is adhered to both sides of a substrate – typically FR4 (flame-retardant fiberglass) but can be made of other materials (FR2, Polyamide, Teflon, etc.)



<https://www.protoexpress.com/kb/pcb-manufacturing-overview/>
<https://www.protoexpress.com/kb/copper-for-pcbs/>

PCB Composition – Solder Mask

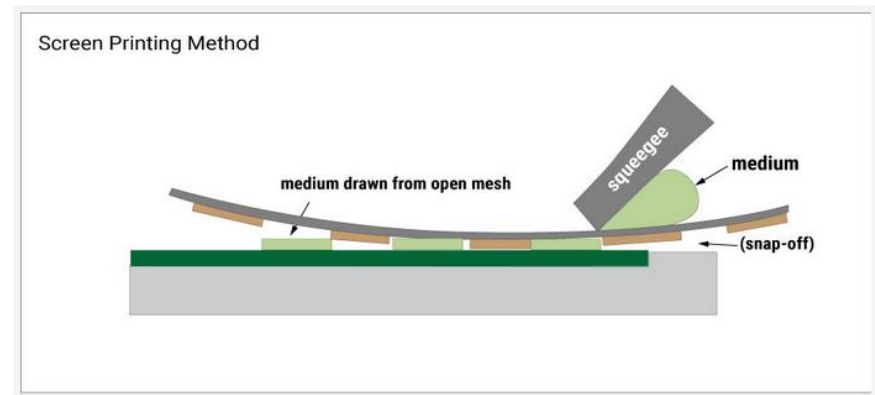
- The solder mask protects regions where solder is not meant to adhere to
- Solder masks result in a hardened layer on the board leaving intended regions exposed



<https://jlcpcb.com/blog/basic-design-of-solder-mask>

PCB Composition – Silkscreen

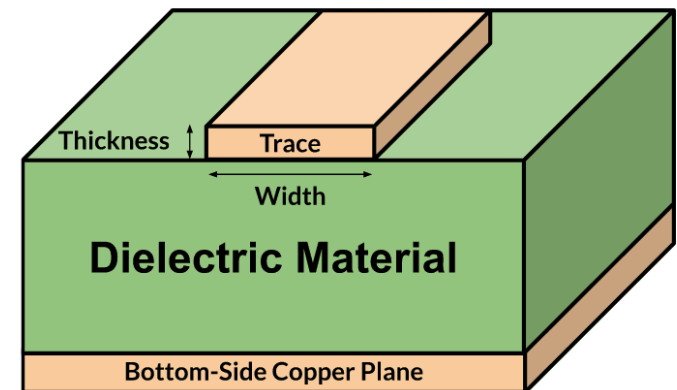
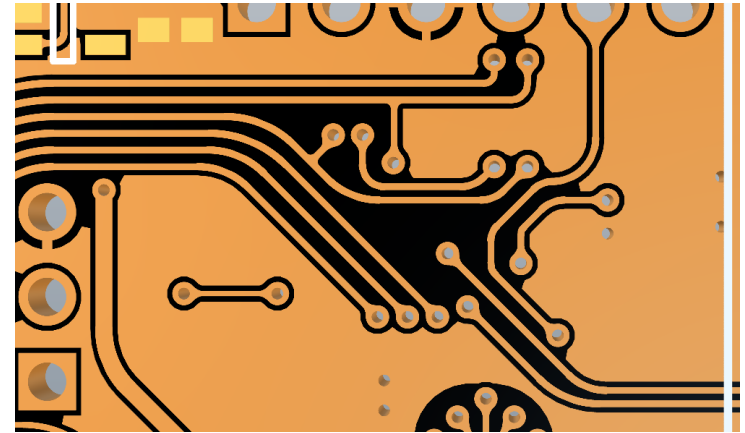
- Silkscreens contain all the text that you will see on the PCB
 - Typically contain component designators
 - Additional notes may be made on the silkscreen
- Silkscreens can only be applied to the solder mask
 - You can send a file to a manufacture that violates this rule, but you may end up with silkscreen on your pads



<https://www.protoexpress.com/kb/silkscreen/>

Traces

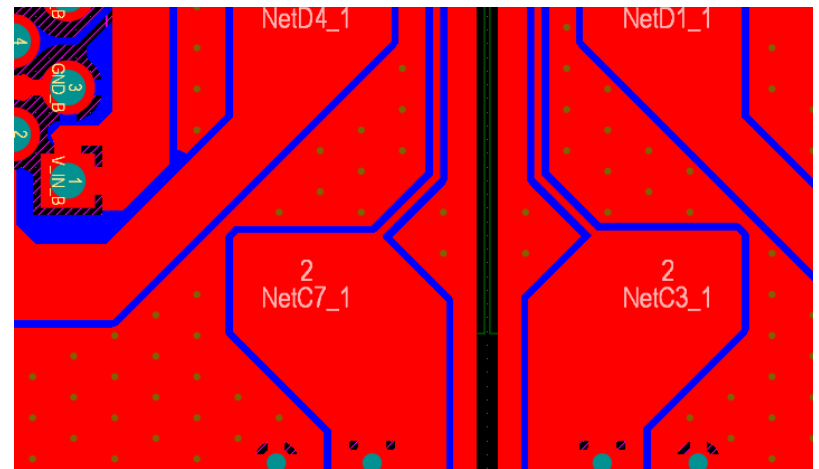
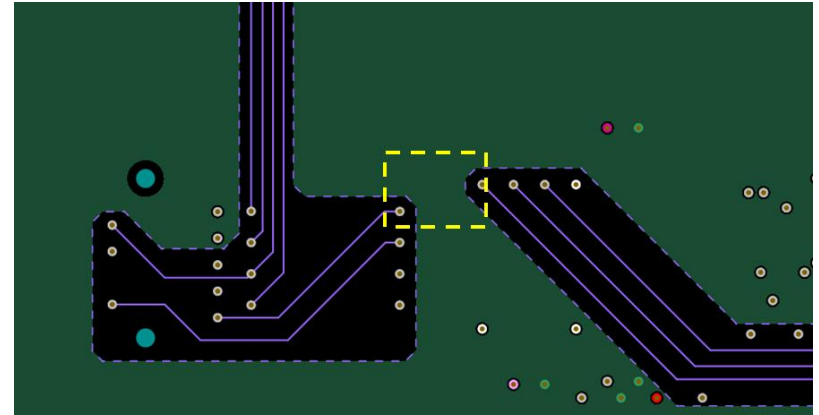
- Traces are the primary way of electrically connecting components to each other
 - Covered in solder mask to prevent short circuits
- Much like wires, traces have resistance and current capacity
 - Capacity dependent on copper weight and width
 - High Current traces must be wider than signal traces



<https://resources.pcb.cadence.com/blog/2024-optimizing-for-pcb-trace-thickness-vs-current-capacity>

Planes

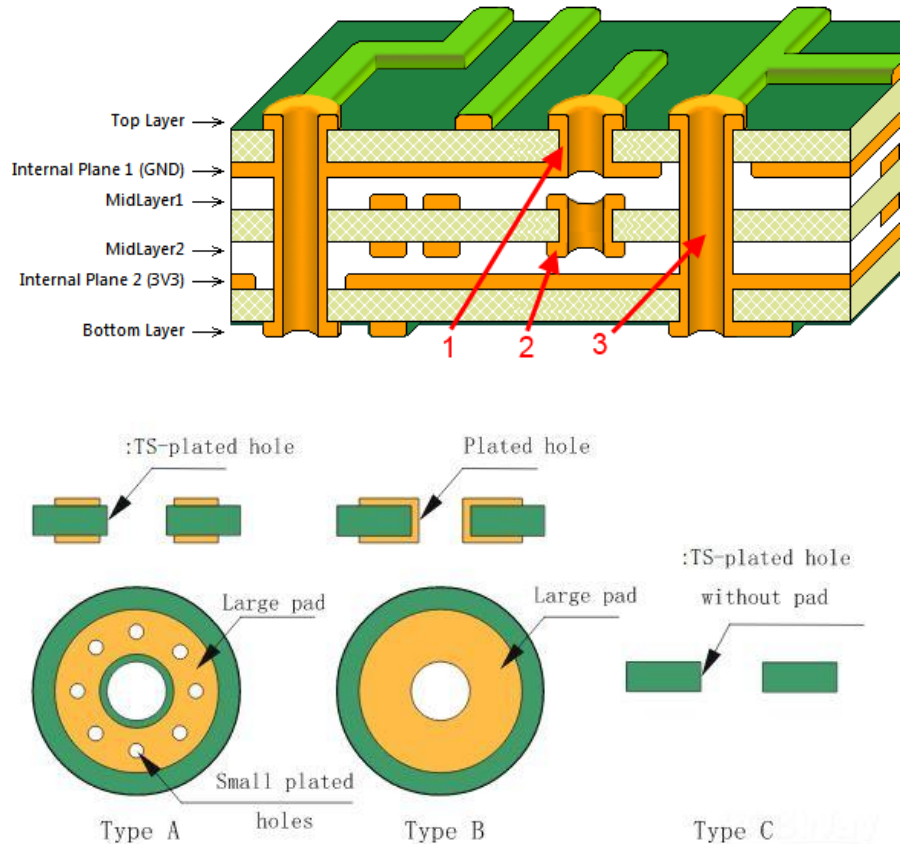
- Planes fill a specified region with copper to electrically connect them
 - Good for GND and Power distribution
 - Good current capacity (alternative to wide traces)
- Planes need special care to remove “necks” and “islands”
- Planes use “thermal reliefs” which need to be checked for capacity



<https://resources.altium.com/p/should-you-route-signals-your-pcb-power-plane>

Vias and Drills

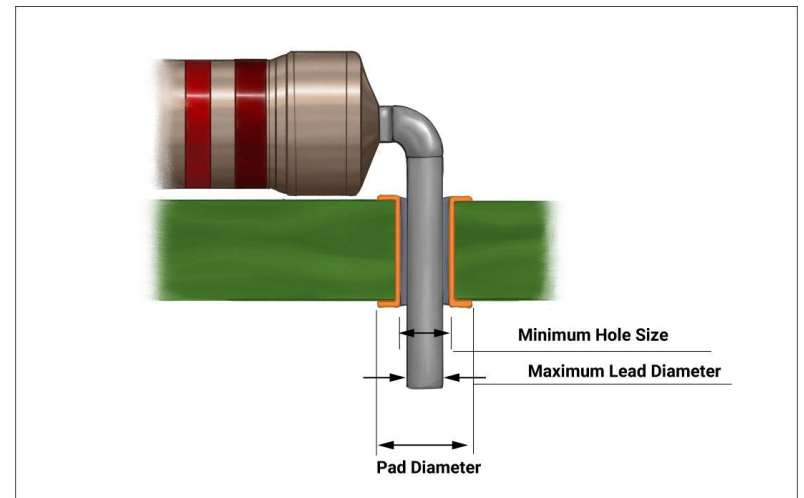
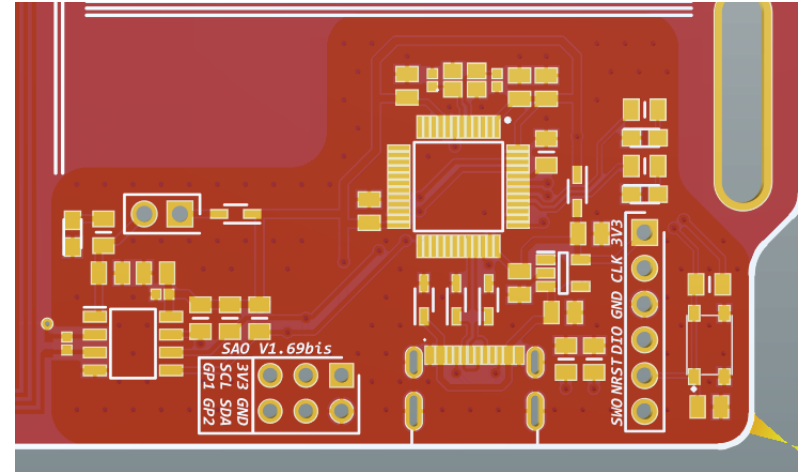
- Vias are holes used to connect between different layers of the PCB
 - Minimize the use of excessive vias when routing
 - Use multiple vias to increase current capacity
- Holes can be plated, non-plated, and have exposed copper if desired
 - Good for mounting points!



<https://www.teachmemicro.com/understanding-pcb-via/>
<https://statics3.seeedstudio.com/fusion/ebook/PCB+DFM+V1.0+.pdf>

Pads

- Pads are exposed copper that can be soldered to
 - Pads will not be covered with the solder mask
 - Pads can either be surface mount or through hole
- Pads dimensions are critical to ensure components solder properly to the PCB



<https://www.pcbasic.com/blog/pcb-pad.html>

Component Footprints

- Footprints are the arrangement of pads that are used to attach components to the PCB
- Footprints must match the component to ensure
- Each component has a unique package, however there are standard sizes.
 - Ex: 0805, 0603, 0402
 - Ex: DIP, TSSOP, QFN

Solder Mask and Solder Paste Recommendations

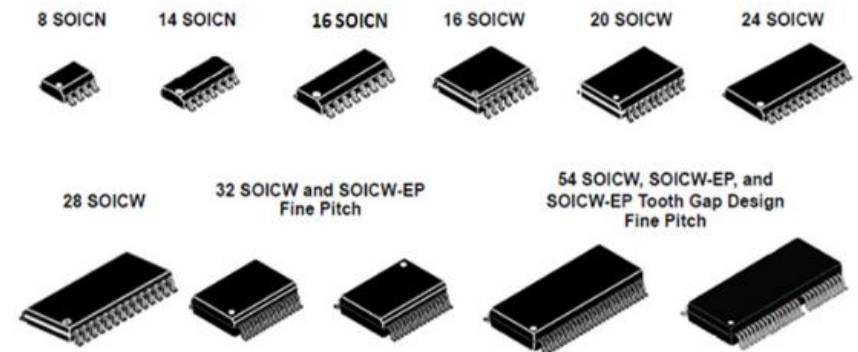
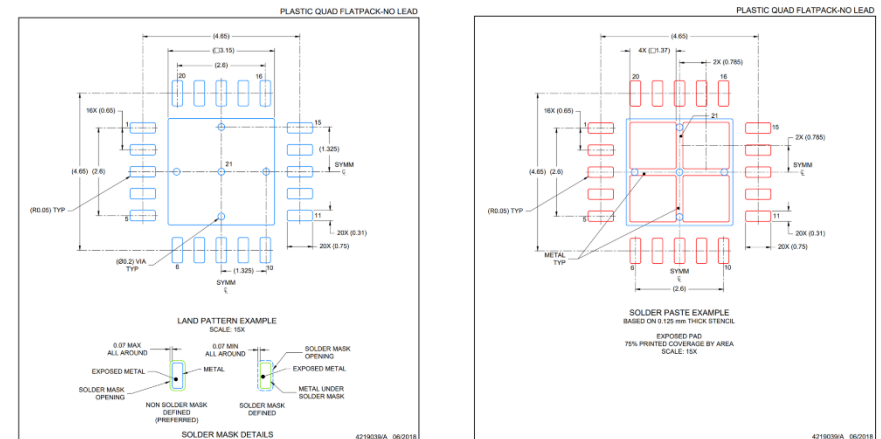
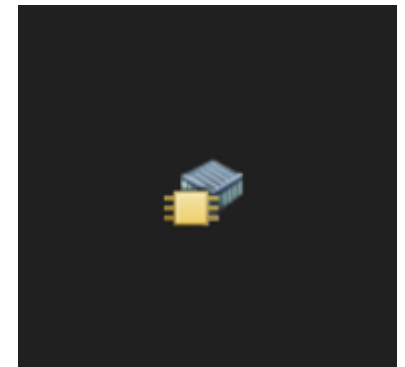
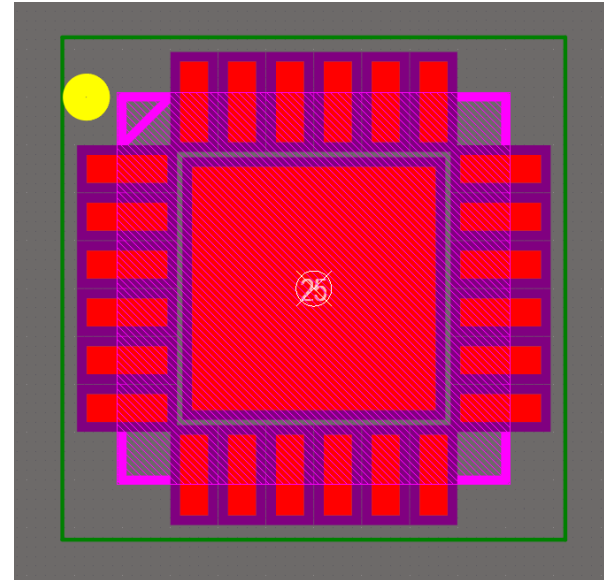


Figure 1. Standard Freescale SOIC Offerings

<https://buildember.com/2024/06/24/component-footprints-what-should-you-include/>
<https://www.nxp.com/docs/en/application-note/AN2409.pdf>

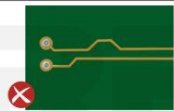
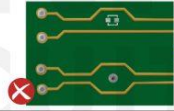
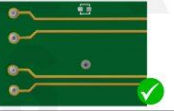
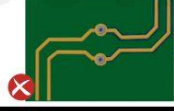
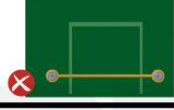
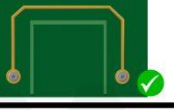
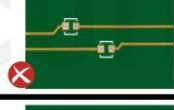
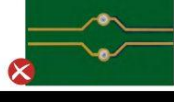
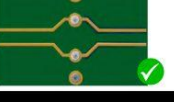
Footprint Libraries

- Very similar to component libraries, although only containing footprints
- Can be utilized across multiple projects
 - Managing a library of very common footprints can save a lot of time
 - SamacSys is also a good tool to help generate footprints quickly
- Altium Footprint Libraries are “.PcbLib”



Layout Tips

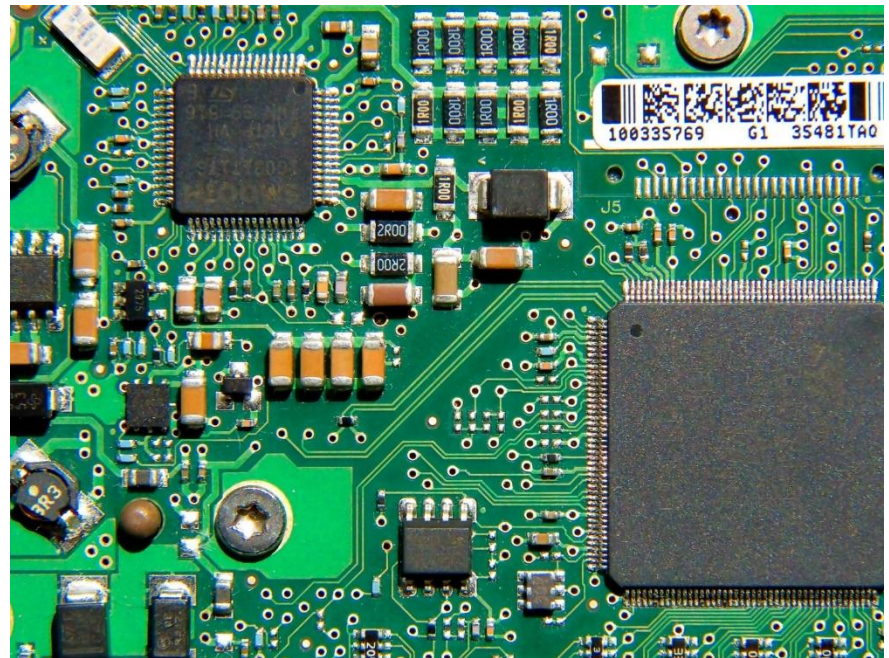
- This is the physical representation of your circuit, placement matters!
- Avoid “weaving” signals with vias
- Avoid running digital traces next to analog signals
- Ensure that traces will be able to handle the current it will carry

SIERRA CIRCUITS		7 PCB Routing Tips to Achieve Controlled Impedance	
S.No	Tips	Layout Dos and Don'ts	
01.	» Route differential pairs symmetrically and keep the signals parallel		
02.	» Avoid placing components and vias between differential lines		
03.	» Add serpentine traces to ensure length matching		
04.	» Do not route high-speed signals over a split plane		
05.	» Incorporate 45° trace bends instead of right-angled bends		
06.	» Maintain symmetry while placing coupling capacitors		
07.	» Place transition vias close to the signal vias		

<https://www.protoexpress.com/blog/best-high-speed-pcb-routing-practices/>

Layout Tips

- Take time to place your components “nicely”
 - Align components
 - Establish a logical flow
- Label components and leave helpful notes for users
 - Ex: Voltage Range, Polarity
- Avoid 90 degree turns in traces
- Place decoupling capacitor close to target devices

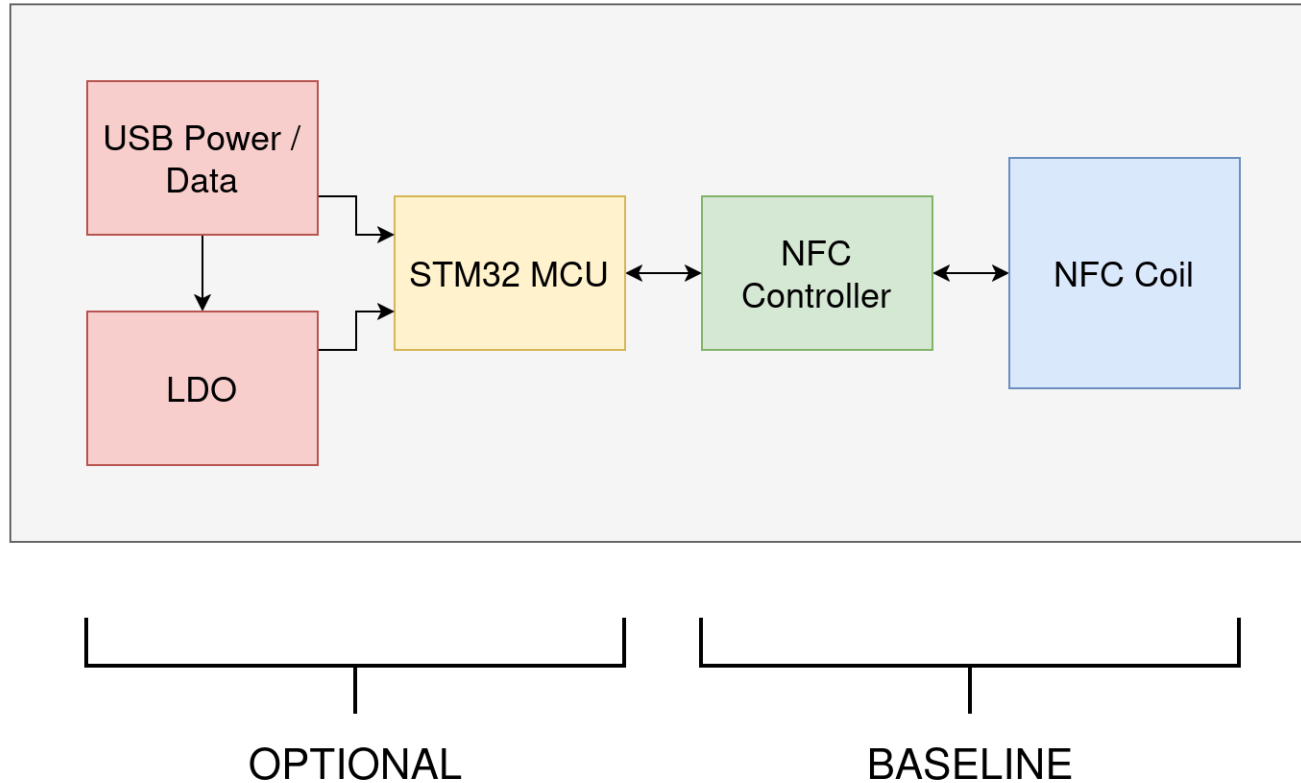


<https://predictabledesigns.com/7-ways-to-quickly-judge-the-quality-of-your-printed-circuit-board-pcb-design/>

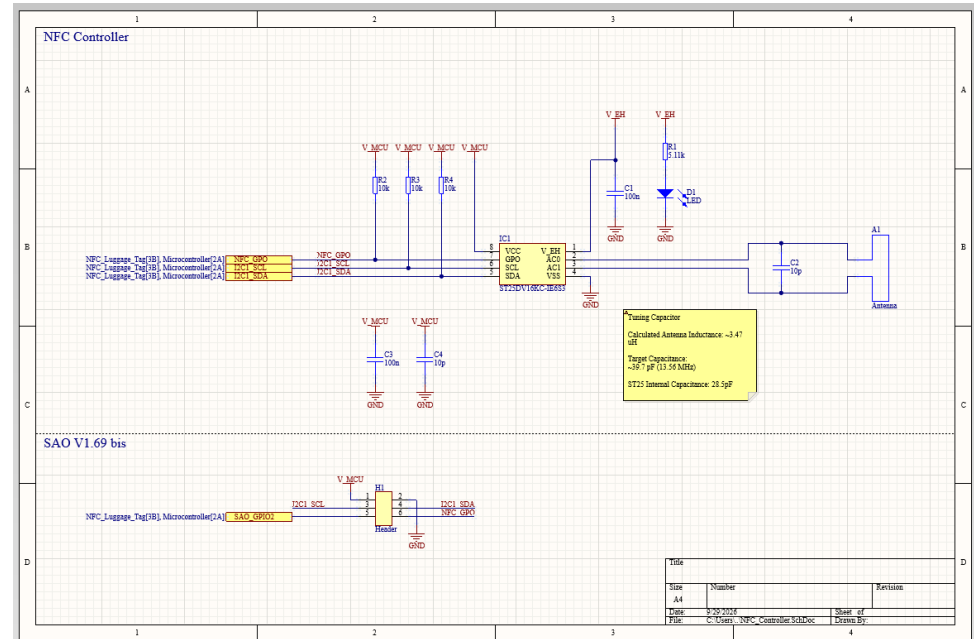
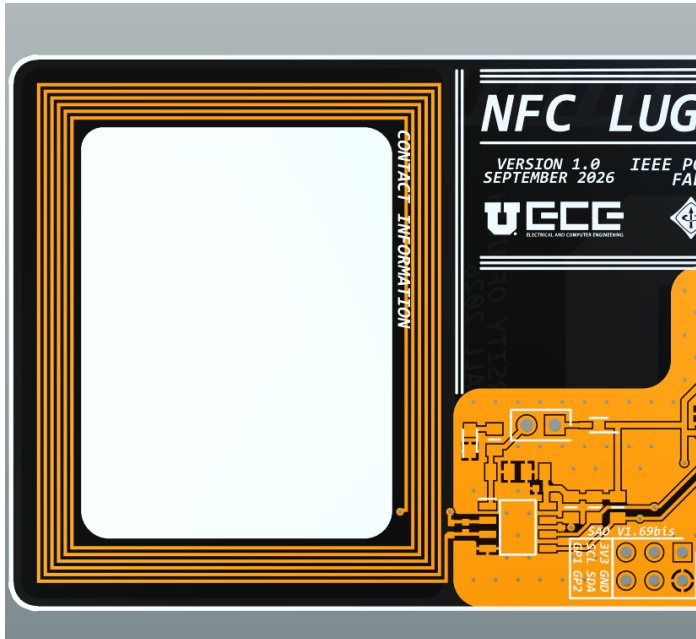
Project Information

- We will be going over the majority of the setup for PCB layout
- If you have not completed the schematic, you may use the reference to proceed
- We highly recommend you follow the reference board dimensions, especially the one highlighted
- Feel free to add more components, it's your own custom board!

Project Block Diagram

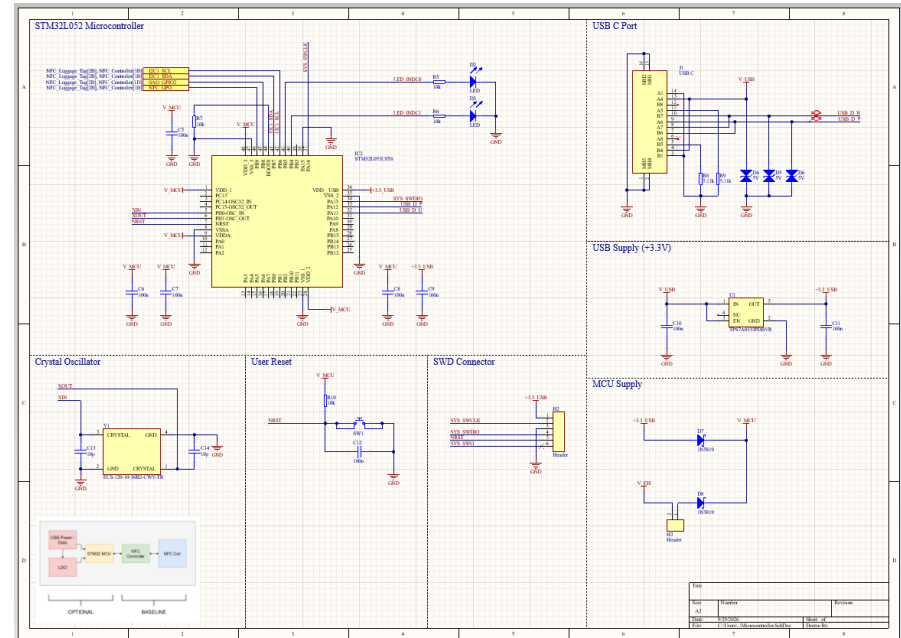
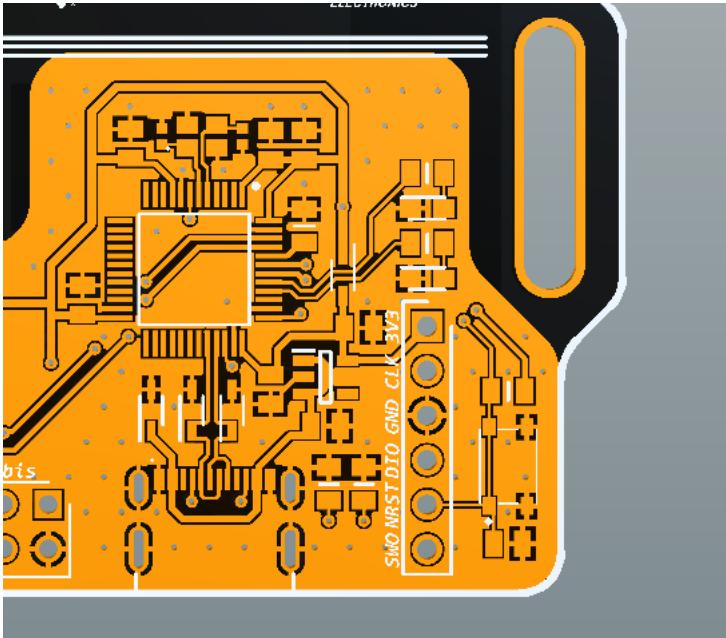


Project Information - Baseline



- Implement the NFC Coil and Coil Controller
- Relatively simple routing, considerations for GND planes and coil placement

Project Information - Optional



- Implement the Microcontroller, USB, and LDO
- More complex considerations:
 - Crystal Placement, Differential Routing, Power Traces

Questions?

Questions?

Download Today's Project Files

Navigate to the workshop GitHub and
download today's files

<https://github.com/IEEE-U-of-U/IEEE-PCB-Workshop-Fall-2026>